

Prime Hypostasis

A Short Reflection on the Idea that Provability + Hypostasis = Classical Semantics

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In the closing section of the Prolegomena, Kant discusses the endeavor of placing metaphysics onto a sound foundation (which is the project of the Critique of Pure Reason) and then setting it up as a proper scientific discipline, capable of generating synthetic judgements a priori.

“To discover the means how the endeavors of the learned may be united in such purpose, I must leave to others. In the meantime, it is [not] my intention to persuade any one merely to follow my propositions, or even to flatter me with the hope that he will do so; but attacks, repetitions limitations, or confirmation, completion, and extension, as the case may be, should be appended. If the matter be but investigated from its foundation, it cannot fail that a system, albeit not my own, shall be erected, that shall be a possession for future generations for which they may have reason to be grateful. It would lead us too far here to show what kind of metaphysics may be expected, when only the principles of criticism have been perfected, and how, because the old false feathers have been pulled out, she need by no means appear poor and reduced to an insignificant figure, but may be in other respects richly and respectably adorned.”¹

We view the endeavor of computational metaphysics as improving the principles of criticism, happening within the tradition of the Critique. The following text is an attempt to synthesize an insight from the twentieth century regarding the relationship between truth and provability with the terminology of the Critique.

¹ Immanuel Kant, Prolegomena to Any Future Metaphysics, trans. Paul Carus (Chicago: Open Court, 1902; Project Gutenberg ebook #52821, released 2016), appendix, <https://www.gutenberg.org/files/52821/52821-h/52821-h.htm>. The bracketed “[not]” has been added to align with the logical sense of the passage and the original German text (“Indessen habe ich nicht die Absicht”), which includes the negation; its omission by Carus seems appears to be an error to us.

Hypostasis is a specific type of representational misunderstanding. It occurs when reason, being confronted with fundamental stabilities in experience, erroneously sublimates them into entities assumed to be somehow stable within themselves.

That is, the stabilities in intuition (*Anschauung*) are turned into something that – supposedly – can be stable in absence of intuition. Those stabilized stabilities are hypostasized space, time, causation, partition and necessity. They are the data format trying to be its own content².

In order for those pseudo-objects to spawn and endure, reason needs to think them in their totality, that is, needs to think them beyond all possible experience. The problem is, of course, that the relational stabilities only exist in the application of functions to a given set of information³. This set of information – that is *raw sensibility* in Kant's lingo – is always specific and itself never a totality. Turning the stabilities into a totality regardless is reason stretching beyond itself. In doing that, the necessary antinomy of pure reason arises, and those hypostasized totalities impose themselves onto the topology of reason as unresolvable knots in thought. On the one hand, those objects are among the most natural parts of our world and *can* be thought locally, on the other hand they strictly *cannot* be thought globally. They simply fall apart. They are functions that are defined for all objects of experience, but undefined outside of that domain. Space as an experiential appearance works as a model of relational properties but space as a self-stable entity can never be fully conceptualized as infinite nor as finite. Time as the representation of a stepwise change along an entropic gradient emerging from a bounded observer being confronted with computational irreducibility is a sensible necessity, but time as a

² i.e. they are a data format necessary for the "cognition-compatibility" of data content, conflated with content itself. Which is not to say that they are not a kind of content itself, those structures, just that they are a different order and type of content that it is a mistake to equivalence with the kind of non-directly-structuring content that is made possible by the data format. I.e.: The data format is to be used on the „objects of experience“ (cognition-compatible structure) but not to be used on the content that is necessary for the cognition compatible structure to arise.

³ The stability is an "operation upon something", and can be abstractly pointed to as "action" independent of the substrate of its application but cannot actually manifest as anything in this abstractly gestured-at form.

self-stable entity breaks apart since it could not have had a beginning, nor could it be without beginning and yet have passed.

Causation works as a local model of information preservation over a variety of state transitions but as a self-stable totality needs a first cause itself. And in a similar way, the local notion of necessity thought as a totality needs to self-justify at some fundamental level. All of this points to the fact that all of those are ill-shaped thoughts. Their objects cannot be confirmed nor disproven by any possible experience and assuming them regardless generates a number of directly contradicting necessities which reveal what Kant calls the *dialectical illusion of pure reason* (*Prolegomena* §52). Therefore the epistemologist must step back in order to formally see that all those non-entities have been generated by the same faulty principle (hypostasis) and stop the application of that principle.

Intermezzo:

Is “hypostasis” just a fancy word for „reification“ or is it a specific type of (that is, a proper subset of) reification? We think it is the latter. It is the subtype of reification that deals with the reification of transcendental ideas, not of objects that might be constructed with their help. Reifying a rainbow is not yet hypostasis, since it reifies a specific appearance but not the rules and forms of all possible appearance (of course those are, by necessity, part of all appearances as well, so there is always some reification of them happening, but hypostasis here is meant to be exclusively aimed at the data-formats themselves, not a specific datum). This is the interesting part because in the former sense, one only reifies something that actually is completely given in experience. Only in hypostasis one reifies that which is not completely given in experience. You can reify a rainbow, stop doing it but still reify the raindrops and the sunlight; you can stop this but still hypostasize space and time in which the de-reified appearance plays out. And there you reify necessary principles, not just tokens or instantiations of those principles. In the systematisches Handlexikon zur KdrV, Heinrich Ratke

catalogues “Hypostase” as “Einen wirklichen Gegenstand außerhalb dem denkenden Subjekte annehmen, was bloß in Gedanken existiert.” (“To take as a real object outside of the thinking subject, what only exists in thought”). This does sound like “plain” reification, not like a strict subset of it. However, the initial section that is mentioned there (A384) is about the hypostatization of “extension” i.e. a property of space itself and movement i.e. a relation-type within the notion of extension. The other sections that are catalogued (e.g. A392, A395, A402, A611) are also mostly about the hypostatization of general principles such as the self, extended matter or ultimate totalities. This is why, for now, we treat hypostasis as a special case of reification⁴.

The entities mentioned so far are just a necessary set of *specific* cases arising from the architecture of human reason and its default settings and representational languages⁵. There *can*, and *have to* be precisely those four types of totalization: totalize the data-formats (space, time), the data-makeup (compound, partition), the data-relations (causation) and the condition

⁴ Another, and in this case hard, criterion for the distinction between reification and hypostasis is this: Hypostasis always deals with a faulty projection of an aspect (be it specific contents or the data formats themselves) of the space of phenomena into the noumenon (thing in itself), while reification can also take place *exclusively within* the space of phenomena. I can reify the notion of a group, a material object, and deconstruct them as ill-defined within the space of appearance while never having claimed that any of them are properties of the thing in itself. I can say “Hey, this is a football game!” and then my opposite might say “The game doesn’t exist, it is 22 humans behaving for 90min according to certain FIFA rules – it is a sequence of state transitions, a sequence of appearances reified into an event that is called a “football game””. And then I could say “Well, 22 humans, that is reified. A human is 37 trillion cells behaving according to certain chemical rules for 2.5 billion seconds, a human is a sequence of state transitions, a sequence of appearances reified into a stable object that is called a “person””. And so on. This chain of reification and deconstruction can go on and on, never leaving the space of appearance. At no point is any hypostatization happening.

⁵ In our case, the hypostatization of the cosmological ideas occurs whenever an entire representational language – usually because the language is mostly used for representing patterns as “outside of” a representing system while at the same time not containing any information about this being a representation – is confused with this very “outside-of-something”-representational content that it is used for to express; that is, the characteristics of the language itself are being assumed as something “outside of” the representing system – in our case, the characteristics of the forms of intuition (space and time) and the categories of understanding (causality, necessity, similarity etc.). Here, in this scenario, hypostasis is the subset of reification that deals with the reification of an entire representational framework, not just with objects that might be constructed within it. Hypostasis can also occur without any specific representation of an „outside of x“, namely, when there are certain all-pervading stabilities in a set of changing appearances, that are then to be assumed to be all-pervading because the appearances are generated by those stabilities (which actually are “just” abstract surfaces for projecting information onto).

for any specific datum (necessity of existence). But there is yet a more general type of hypostatization to be found in all possible kinds of human reason, one that *underlies* all the specifics of the protagonist-entities of the antinomies of pure reason. It is the result of applying the same faulty principle to the notion of truth⁶ itself.

The hypostatization of truth – although it constitutes what is probably the greatest metaphysical sacrilege – is of an almost perfidious subtlety, since truth seemingly manifests itself in all types of judgements and is furthermore to be encountered – and needed – anywhere one happens to look, indeed being the most stable structural principle of all, one that is common to all concepts and all critiques. And, as is the case with space/time/causation/partition/necessity, truth needs to be recognized as ideal (in the Kantian sense, not in the Platonic sense) and not part of the data. Truth, in its various definitions and demarcations, is the condition of the possibility of structure in concepts and of all theories. It even is the condition of the possibility of all languages. It is reason's sorting-tool which allows reason to build the relations in which it operates in the first place.

Truth is not only encountered but also *discovered*⁷ everywhere we look, as if it had already been there all along, which leads us to suspect its existence beyond all things. Even though we might say „truth is a property of sentences“, we still project truth into the thing in itself.

⁶ The proper intension of truth (if one supposes it to be a meaningful notion beyond constructive mathematics in the first place) seems to be unclear to us. But there *are* extensional instances which are worked with and also referred to by the term „truth“, and a large number of those instances *are* hypostatized. Those are what we address. That is all we need for our argument at hand. Examples of instances of this kind are: Assigning a truth value explicitly or implicitly to an undecidable function, calling upon a correspondence between a propositional statement and some non-constructive „actuality“ beyond all observation, making any claim about the structure of some non-constructive universe, claiming the existence of a true but unprovable statement, any kind of mathematical operation in which an infinity is applied as an intermediate step that then results in a final value.

⁷ The *discovered* relates to the metamathematical debates on whether mathematics is invented or discovered – a question people often ask at some point. Here we intend to express that truth, as reason's sorting tool, is to be encountered everywhere (in the sense of being constructed, or implemented by reason), but also is encountered sometimes in a way that is more akin to discovery e.g. the Church-Turing thesis as something that seems to be a fact about formality itself that holds independent of what kind of sorting filter might be employed by a given system.

Truth seems to be, even though it can be defined, recognized, computed, explicated and checked, independent of all those endeavors. A constructive notion of truth, which excludes the law of excluded middle, looks interesting at first, second and nth sight but nevertheless seems reductionist, merely descriptive, missing out on actuality, missing the mark of the deep, the *true* truth beyond.

Similar to the case of dreaming a partial awakening from the pre-critical dogmatic slumber *within* this very slumber by realizing that color, smell and sound are not part of the world, but still believing that space, time and entropy are, we witness the dreaming of the awakening from the pre-computational dogmatic slumber *within* this very slumber, whenever people claim that „truth is a property of sentences“ but at the same time assert that the principle of the construction of all true sentences – that is, provability, the possibility of a finite, stepwise, gapless synthesis/derivation⁸ of a well-formed statement, starting with the axioms of a given formal system – is less encompassing than truth itself. But if truth is a property of sentences then it cannot exceed the powers of the construction principle of all true sentences⁹; the truth-property must be assignable to statements as formal objects by some formally specifiable procedure that operates on them. (And any such procedure within a formal system is a proof protocol, and the truths it reaches are precisely those it proves.)

⁸ Provability is formal derivability within a system (Hilbert/Gödel). By the definition of a formal system, provability as formal derivability is a computable process, and the Church-Turing thesis states that there is no notion of “formally specifiable procedure” that exceeds computation. Turing further added that the general question of whether an arbitrary statement is provable in a given system is itself undecidable, that is, that there exists no algorithm that can determine for every sentence whether or not a derivation of it exists.

⁹ For “real” (finitary) statements whose quantifiers range over bounded domains, truth is decidable via finite computation, since one can simply check all cases. For statements involving unbounded quantifiers, where such exhaustive checking is impossible, truth in the classical sense must be assigned by other means: the formal system must be consistent, and ideal (infinitary/classical) methods must be conservative over real ones (they prove no new real truths that could not in principle be established by finitary means). Hilbert’s program conjectured that a finitary consistency proof would justify all classical “truths” as reliably derivable, so that the question of whether truth and provability could be decoupled did not arise in its modern form. This is the question that Gödel would quickly and decisively force open.

In that, accepting incompleteness (the existence of a true but unprovable statement within a given system) is to admit into the system itself a non-formalizable notion of truth, that is not a formally assignable property of sentences, which is to suspend the system, so that it is groundless and levitating, from a hook that is attached to nothing. Here we realize, that asserting that truth is sentential, while at the same time asserting that provability and truth are distinct, that is, that truth exceeds all sentential apparatus – instead of treating truth as the hypostatization of provability – while simultaneously failing to see the significance of the resulting antinomies spawning left and right, is a sign that we are slumbering peacefully through a thunderstorm.

As of today, finitist stances seem to draw quite some skepticism. We suspect this is because when thinking about constructivist mathematics and its implications, one still consciously or unconsciously applies the notion of finiteness to the thing in itself, where the notion does not make a whole lot of sense. The notion of infinity doesn't either, but at least this one does not consist of a clearly defined set of decidable functions, constraining the nature of the thing in itself in a weirdly specific way. But finiteness is nothing that can be used on the things in themselves. It is to be used in the space of phenomena. Finiteness is not a claim beyond the realm of phenomena in the first place, infinity is. This is the difference. Classical mathematics and constructive mathematics, in so far as foundational debates are concerned, are not competing for a claim on the things in themselves even if many people involved in the debates might think they do. They compete for the proper structure of epistemology and, in that, reason itself; and here we see that only hypostasis can give birth to classical mathematics, it is pre-critical in a foundational sense, no matter how well it works as a practical approximation or as a powerful theoretical formalism in so many cases. Constructive mathematics in turn, is critical in a foundational sense. It is the result of reason critiquing not just the transcendental

ideas, arising from the forms of intuition and the categories of understanding unified by apperception, but the complete set of languages which it can use to think anything.

Even though all of this has already been said and done in other words, nowadays, within a large number of philosophical discourse spaces, it is considered good form to claim that the „whereof one cannot speak“ i.e. the infinity/uncomputability of deep reality, unfortunately cannot be grasped within the limited framework of the human mind and formality itself. But in this very case there is, once again, the speaking of that of which one cannot speak – that is, one imagines doing so and elevates oneself above the limits of knowledge in order to invent untenable metaphysical entities, which one then places above what is possible, thereby viewing the possible as reductionist. A computational-constructivist stance is often perceived as claiming, in impermissible simplification, to penetrate everything, stubbornly excluding what does not fit into this rigid world view. However, the actual problem lies in the fact that if one believes that their mind allows them to overcome the limits of the formally possible, they are mistaking dogmatic etiquette for critical thinking – but the foundational formalism in which philosophy is done cannot contain hypostatized elements.

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